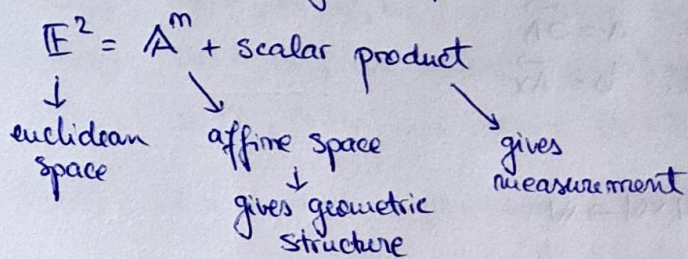


$\mathbb{E} \rightarrow$ Euclidean space

- Euclid's postulates ~ 300 BCE
- Foundations of geometry: Hilbert ~ 1899



- Lean mathlib $\rightarrow$ language (~ 2013)

Geometric vectors

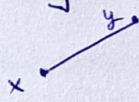
- oriented segment: $[AB]$
 $A \rightarrow B$

Def: the (free) vector represented by A and B:

$$B - A = \vec{AB}$$

- equipolency:

Def: $[xy] \sim [AB]$ (equipotent) if $\begin{cases} [AX], [BY] \text{ have the same midpoints.} \\ [xy], [AB] \text{ oriented segments.} \end{cases}$



$\rightarrow$ if A, B, x, y noncollinear $\Rightarrow$ equipolency $(\Rightarrow)$ $AB \parallel xy$ parallelogram.

$\rightarrow$ equipotent = same magnitude + direction.

$\rightarrow$ equipotence relation = equivalence relation

$\rightarrow$ vectors = equivalence classes of the equipotence relation.

- $\vec{AB}$: the vector containing the oriented segment $[AB]$
 $\Rightarrow [AB]$ "is representative" of the vector $\vec{AB}$.

- $\vec{AB} = \vec{CD} \Rightarrow [AB], [CD]$ define the same vector.

The set of all vectors defined with points in $\mathbb{E}$:

$$V = \{ \vec{AB} : \text{where } A, B \text{ are points in } \mathbb{E} \} = \mathbb{E} \times \mathbb{E} / \sim$$

$\hookrightarrow$ (real) vector space (V with vector addition and scalar multiplication)

$$\vec{AA} = \vec{0}$$

length of a vector: $|\vec{AB}| = |B-A|$.

$$\vec{BA} = -\vec{AB}$$

Addition of vectors:

$$1) \begin{cases} a, b \text{ vectors.} \\ 0 - \text{fixed point} \end{cases} \Rightarrow \begin{cases} \exists! A \text{ s.t. } a = \vec{OA} \\ \exists! X \text{ s.t. } b = \vec{AX} \end{cases}$$

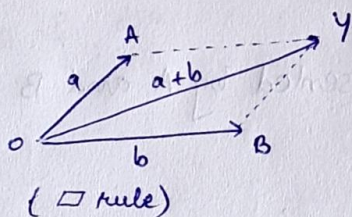
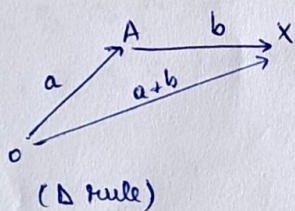
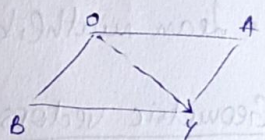
$$a+b = \vec{OA} + \vec{AX} = \vec{OX}$$

$\Rightarrow$ defines a map: $\square + \square : \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{V}$

$$2) 0 - \text{fixed point} \Rightarrow \exists! A, \exists! B \text{ s.t. } \begin{cases} a = \vec{OA} \\ b = \vec{OB} \end{cases}$$

$$3) 0, A, B \Rightarrow \exists! Y \text{ s.t. } OAYB \text{ parallelogram.}$$

$$\Rightarrow \underline{X=Y} \Rightarrow a+b = \vec{OY} = \vec{OX}$$



Multiplication of vectors with scalars:

$$\begin{cases} a = \vec{OA} \\ x \in \mathbb{R} \end{cases} \Rightarrow \underline{x > 0} \Rightarrow \exists! X \text{ s.t. } |OX| = x \cdot |OA|$$

$$\Rightarrow x \cdot a = xa$$

$\Rightarrow$ defines a map $\square \cdot \square : \mathbb{R} \times \mathbb{V} \rightarrow \mathbb{V}$

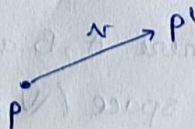
$$x \cdot a = \begin{cases} \vec{OX}, & \text{for } a \neq \vec{0}, x > 0 \text{ and } x \text{ as above} \\ -(|x|a), & \text{for } a \neq \vec{0}, x < 0 \\ \vec{0}, & \text{for } a = \vec{0} \text{ or } x = 0 \end{cases}$$

Affine space

• if you fix a point O , $r_O(P) = \vec{OP} \Rightarrow \text{pts} \xleftrightarrow{1:1} \text{vectors.}$

$\downarrow$
bijection

• if $v \in \mathbb{V}$, P point, $v+P = P'$



Affine space = set of points to which we attach a vector space V and

$\square - \square : A \times A \rightarrow V$ } maps, subject to the axioms:
 $\square + \square : V \times A \rightarrow A$ }

(AS₁): $(\forall) P$, we have $\vec{0} + P = P$.

(AS₂): $(\forall) P, (\forall) v, w$, we have $(v+w)+P = v+(w+P)$.

(AS₃): $(\forall) A, B$, we have $(A-B)+B = A$.

(AS₄): $(\forall) P, (\forall) v$, we have $(v+P)-P = v$.

$\square - \square : A \times A \rightarrow V$ produces a vector from 2 points.
 $(A-B = \vec{AB})$

$\square + \square : V \times A \rightarrow A$ produces a point from a vector and a point
 = "translation"
 $(v + B = \vec{AB})$
 $\uparrow$
 (A)

Euclidean space

Def: Theorem: E is an affine space of dimension ≥ 3 .

Direction space of A .

$D(A)$: the vector space V attached to the affine space A .

$\dim(A) = \dim(D(A)) = \dim V$.

$A^m = m$ -dimensional affine space.

Affine subspace

Def: Let A affine space. A subset $S \subseteq A$ is called affine subspace if $\exists$ a point P and a vector subspace $U \subseteq D(A)$ s.t.

$$S = U + P = \{v + P, v \in U\}$$

$\Rightarrow U$ unique subspace, denoted by $D(S)$. ("direction subspace of S ")

$\dim S = \dim D(S)$.

• A point is an affine subspace of dimension 0

(S is a point if $\dim U = 0$)

• A line is an affine subspace of dimension 1

(S is a line if $\dim U = 1$)

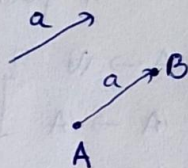
• S is a plane if $\dim U = 2$

• S is a hyperplane if $\dim U = \dim A^m - 1$.

affine space becomes
 ambient space if we
 fix the origin.

"Act on a point"

vector a } $\Rightarrow \exists ! B$ s.t. $a = \vec{AB}$
point A



"the vector a is moving the point A to the point B "

(vectors act on points)

$\hookrightarrow$ this interaction $\rightarrow$ data structure: affine space.

Cartesian frames

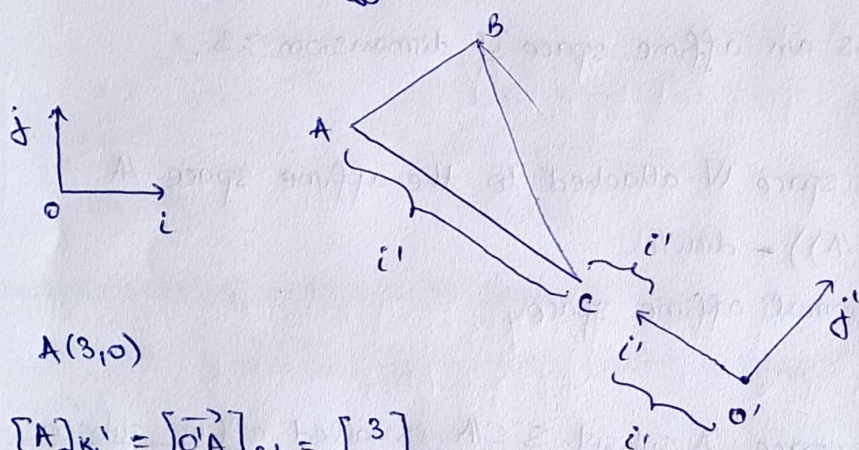
$\rightarrow$ René Descartes ~ 1637

$\rightarrow$ ways of giving coordinates

Def: A frame of A^2 is a pair $K = (O, B)$

$\begin{pmatrix} i, j \\ (i_1, \dots, i_m) \end{pmatrix} \leftarrow ?$

$$\text{Points} \xleftrightarrow[\text{O}]{1:1} V \xleftrightarrow[B]{1:1} \mathbb{R}$$



$A(3, 0)$

$$[A]_{K'} = [\vec{O'A}]_{B'} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$[A]_{K'} = 3i'$$

$$[B]_K = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$[B]_{K'} = \vec{O'A} + \vec{AB} = 3i' + j' \Rightarrow \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Changing frames

$\rightarrow$ recall change of basis (matrix)

V vector space;

$$[w]_{\mathcal{E}} = M_{\mathcal{E}, \mathcal{F}} \cdot [v]_{\mathcal{F}}$$

Let $\mathcal{E}, \mathcal{F}$ be 2 bases of V . Then the change of basis matrix from

$$\mathcal{F} (f_1, \dots, f_m) \text{ to } \mathcal{E} (e_1, \dots, e_m) \text{ is } M_{\mathcal{E}, \mathcal{F}} = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ [f_1]_{\mathcal{E}} & [f_2]_{\mathcal{E}} & \dots & [f_m]_{\mathcal{E}} \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

Theorem 1.10.

Let $K=(O, B)$ and $K'=(O', B')$ be 2 frames on A^m .

For any point $P \in A^m$ we have:

$$(*) \quad [P]_{K'} = M_{B'B} \cdot ([P]_K - [O']_K) = M_{B'B} \cdot ([P]_K - [O']_K) = M_{B'B} \cdot [P]_K + [O]_{K'}$$

Proof.

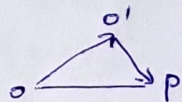
Since the coordinates of a point are the components of its position vector, $(*)$ is equivalent to:

$$\begin{aligned} (*) \Leftrightarrow [\vec{O'P}]_{B'} &= M_{B'B} ([\vec{OP}]_B - [\vec{OO'}]_B) \\ &= M_{B'B} \cdot [\vec{O'P}]_B \end{aligned}$$

$$M_{B'B} \cdot (-[O']_K) = [\vec{O'O}]_{B'} = [O]_{K'}$$

$$\begin{aligned} &= \\ &[\vec{O'O}]_B \end{aligned}$$

$$M_{B'B} \cdot [\vec{OO'}]_B = [\vec{OO'}]_{B'} = -[\vec{O'O}]_{B'}$$



$$OO' + O'P = OP$$

$$OP - OO' = \underline{O'P}$$